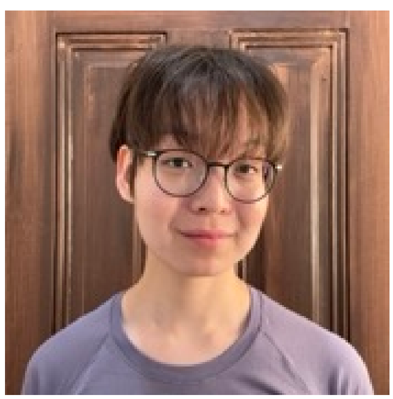
# Statistics 151a: Linear models

**UC Berkeley, Spring 2026**

## Instructors

**Instructor:** Ryan Giordano  Office hour location: 389 Evans Hall Office hours: Thur 11am-12pm, Tues 4pm-5pm `rgiordano@berkeley.edu` pronouns: He / him

**GSI**: Danat Dusinbekov  Office hour location: 428 Evans Hall Office hours: Wed 1pm-3pm, Mon 12pm-2pm `danat@berkeley.edu` pronouns: He / him

! Important

Please reach out to Danat if you need to be manually added to BCourses.

## Schedule

Lectures will be held on Monday, Wednesday, and Friday from 9–10am in Davis 534. The first lecture will be on Jan 21st, and the last will be on Friday May 1st.

Labs will be held on Wednesdays from 11 am – 1 pm and 3–5 pm in Evans 340.

(Link to official course calendar.)

## Materials

Unelss otherwise noted, the primary materials for the course are the lecture notes, which will be posted to the course website in advance of class. The following textbooks are useful supplementary texts and are all either freely available online or available electronically through the UC Berkeley library:

- SMT: Statistical Models: Theory and Practice Freedman
- LME: Linear Models and Extensions Ding
- ISL: An Introduction to Statistical Learning James, Witten, Hastie, Tibshirani
- VDS: Veridical Data Science Yu, Barter
- ETM: Econometric Theory and Methods Davidson, MacKinnon

Other books of interest to the class are

- FPP: Statistics Freedman, Pisami, Purves
- RDS: R for Data Science, Wickham, Grolemund
- ROS: Regression and other Stories Gelman, Hill, Vehtari

### Linear algebra review materials

Basic linear algebra is a serious prerequisite for this course. You can find a summary of useful review materials here.

### R programming review materials

This course will be conducting in the `R` programming language, and basic proficiency is assumed. Here are some useful review materials:

- Berkeley SCF R bootcamp materials
- Berkeley D-Lab R fundamentals workshop

## Course Calendar

**The following schedule is subject to change.**

Table 1: Calendar (tentative)

| Date | Day | Note | Unit | Topic | Assignment |
|---|---|---|---|---|---|
| Jan 21 | Wednesday | Lecture | Introduction | Class policies | Lab 0 |
| Jan 23 | Friday | Lecture | | Real-world questions | |
| Jan 26 | Monday | Lecture | | Inference and probability | |
| Jan 28 | Wednesday | Lecture | | Prediction and uncertainty | Lab 1 |
| Jan 30 | Friday | Lecture | | Linear algebra review | HW 0 |
| Feb 2 | Monday | Lecture | Unit 1: Multilinear regression | Simple linear regression | |
| Feb 4 | Wednesday | Lecture | | Multilinear regression | Lab (review) |
| Feb 6 | Friday | Lecture | | ——— | Quiz 0 |
| Feb 9 | Monday | Lecture | | One-hot encodings | |
| Feb 11 | Wednesday | Lecture | | Interactions | Lab 2 |
| Feb 13 | Friday | Lecture | | Transforming the response | |
| Feb 16 | Monday | Administrative holiday | | | |
| Feb 18 | Wednesday | Lecture | Unit 2: Testing and inference | The multivariate normal | Lab 3 |
| Feb 20 | Friday | Lecture | | Asymptotics and consistency | HW 1 |
| Feb 23 | Monday | Lecture | | Asymptotics and consistency | |
| Feb 25 | Wednesday | Lecture | | Confidence intervals | Lab (review) |
| Feb 27 | Friday | Lecture | | ——— | Quiz 1 |

| Date | Day | Note | Unit | Topic | Assignment |
|---|---|---|---|---|---|
| Mar 2 | Monday | Lecture | | Heteroskedasticity and grouping | |
| Mar 4 | Wednesday | Lecture | | Test statistics under normality | Lab 5 |
| Mar 6 | Friday | Lecture | | Model selection and F-tests | |
| Mar 9 | Monday | Lecture | Unit 3: Regularization and machine learning | Splines and basis expansions | |
| Mar 11 | Wednesday | Lecture | | Ridge regression | Lab 6 |
| Mar 13 | Friday | Lecture | | Lasso regression | HW 2 |
| Mar 16 | Monday | Lecture | | Cross validation and model selection | |
| Mar 18 | Wednesday | Lecture | | Conformal intervals | Lab (review) |
| Mar 20 | Friday | Lecture | | ——— | Quiz 2 |
| Mar 23 | Monday | Spring break | | ——— | |
| Mar 24 | Tuesday | Spring break | | ——— | |
| Mar 25 | Wednesday | Spring break | | ——— | |
| Mar 26 | Thursday | Spring break | | ——— | |
| Mar 27 | Friday | Spring break | | ——— | |
| Mar 30 | Monday | Lecture | Unit 4: Criticism and diagnostics | Outliers and leverage | |
| Apr 1 | Wednesday | Lecture | | The influence function | Lab 7 |
| Apr 3 | Friday | Lecture | | Regression to the mean | HW 3 |
| Apr 6 | Monday | Lecture | | The FWL theorem | |
| Apr 8 | Wednesday | Lecture | | Omitted variable bias | Lab (review) |
| Apr 10 | Friday | Lecture | | ——— | Quiz 3 |
| Apr 13 | Monday | Lecture | Unit 5: Generalizations | Logistic regression | |
| Apr 15 | Wednesday | Lecture | | Poisson regression | Lab 8 |
| Apr 17 | Friday | Lecture | | Nonlinear least squares | HW 4 |
| Apr 20 | Monday | Lecture | | Random effects models | |

| Date | Day | Note | Unit | Topic | Assignment |
|---|---|---|---|---|---|
| Apr 22 | Wednesday | Lecture | | Hierarchial modeling | Lab (review) |
| Apr 24 | Friday | Lecture | | Hierarchial modeling | Quiz 4 |
| Apr 27 | Monday | Lecture | Unit 6: Special topic | TBD | |
| Apr 29 | Wednesday | Lecture | | TBD | Lab (projects) |
| May 1 | Friday | Lecture | | TBD | HW 5 |
| May 4 | Monday | Lecture | Final projects | | |
| May 6 | Wednesday | Lecture | | | Lab (projects) |
| May 8 | Friday | Lecture | | | |

Unelss otherwise noted, the primary materials for the course are the lecture notes, which will be posted to the course website in advance of class. The following textbooks are useful supplementary texts and are all either freely available online or available electronically through the UC Berkeley library:

- SMT: Statistical Models: Theory and Practice Freedman
- LME: Linear Models and Extensions Ding
- ISL: An Introduction to Statistical Learning James, Witten, Hastie, Tibshirani
- VDS: Veridical Data Science Yu, Barter
- ETM: Econometric Theory and Methods Davidson, MacKinnon

Other books of interest to the class are

- FPP: Statistics Freedman, Pisami, Purves
- RDS: R for Data Science, Wickham, Grolemund
- ROS: Regression and other Stories Gelman, Hill, Vehtari